

NEOLABS

SECURITY LABS | SOC LEVEL 1

Week 02 - The Ghost Login

Weekly Learning Pack | Learn + Connect + Operate

Programme: NeoLabs x Renaissance Innovation Labs Cybersecurity Internship

Week 02

Version: 1.0

Classification: Student Training Material - Authorised Synthetic Use Only

Authorised synthetic training use only.

Why this week matters

A suspicious-login report is one of the most common ways a security investigation begins. A user may report an unfamiliar sign-in, a strange notification, a failed login they do not remember, or activity at an unexpected time. The analyst's job is not to immediately label every unusual event as an attack. The analyst must collect evidence, understand normal behaviour, build a timeline and decide what the evidence supports.

Week 2 introduces the SOC habit that will be used throughout the internship: **baseline first, evidence second, conclusion last**.

Learning outcomes

By the end of this pack you should be able to:

- explain the difference between authentication, authorisation and session activity;
- recognise useful fields in authentication telemetry;
- distinguish ordinary login mistakes from suspicious patterns;
- build a chronological authentication timeline;
- correlate multiple events without relying on one alert alone;
- record evidence in a way another analyst can reproduce;
- decide when an event is benign, suspicious or a confirmed incident inside the lab story;
- propose a practical Wazuh detection improvement;
- validate that a fixed scenario no longer produces the same unsafe behaviour.

1. Authentication telemetry

Authentication telemetry is the evidence produced when a system verifies identity. In the VCC lab this may include application logs, API events and Wazuh-ingested telemetry. Useful fields commonly include:

Field	Why it matters
Event time	Establishes sequence and duration
User/account	Identifies the synthetic identity involved
Result	Successful, failed, blocked or challenged login
Source address	Helps compare origin patterns inside the authorised lab
User agent/device	Helps distinguish expected and unexpected client behaviour
Event type	Tells you whether the record represents login, logout, failure or another auth action
Request/session identifier	Helps correlate related events
Failure reason	May distinguish a typo, disabled account or controlled suspicious activity

An individual field rarely proves the whole story. Analysts gain confidence by correlating several fields across time.

Analyst note: An unfamiliar source address is an indicator to investigate, not automatic proof of compromise. Context still matters.

2. Baseline before anomaly

A baseline is a picture of expected behaviour. Before investigating the suspicious period, review normal synthetic login activity for your assigned pod.

Ask:

- Which accounts normally authenticate?
- What does a normal successful login look like?
- What does a normal password mistake look like?
- Are there predictable retries?
- Which fields remain stable across ordinary activity?
- Which events are generated by the application itself versus a learner action?

When you understand the baseline, anomalies become easier to explain.

3. Patterns that deserve attention

Inside this controlled exercise, investigate patterns such as:

- repeated failed attempts followed by a success;
- an account appearing from an unexpected lab source;
- several synthetic accounts showing a similar pattern in a short period;
- success events with no expected preceding user behaviour;
- unusual timing compared with the baseline window;
- a mismatch between the user's report and the event sequence.

Do not assume all of these are malicious. Your report should state what the evidence shows and what remains uncertain.

4. Build a timeline

A useful authentication timeline is concise and chronological.

Example structure:

Time (UTC)	Account	Event	Source	Analyst interpretation	Evidence reference
10:03:11	learner-a	Login failed	lab-source	Possible typo	EV-01
10:03:19	learner-a	Login success	same source	Consistent with retry	EV-02
10:18:42	learner-b	Login success	different source	Requires correlation	EV-03

Use the timestamps from the telemetry. Do not rewrite a timeline from memory after the investigation.

Evidence requirement: Every important conclusion in your final report should point back to a screenshot, event identifier, redacted log excerpt or other approved evidence reference.

5. Wazuh investigation workflow

After your pod connection is active:

```
python3 tools/neolabs.py status
python3 tools/neolabs.py pod info
```

Your local Wazuh environment should receive only the telemetry associated with your server-assigned pod.

A beginner-friendly investigation sequence is:

- 1 Confirm the assigned time window from the GitHub Issue.
- 2 Review the normal baseline period.
- 3 Filter for authentication-related events.
- 4 Identify the synthetic account(s) mentioned in the exercise.
- 5 Compare failed and successful events.
- 6 Correlate timestamps, source fields, user agent/device fields and event identifiers.
- 7 Add relevant evidence to your register.
- 8 Build the timeline.
- 9 State the most likely explanation and your confidence.
- 10 Escalate only when the evidence meets the exercise threshold.

6. Avoid common analyst mistakes

Alert equals incident

An alert is a signal to investigate. It is not the conclusion.

One strange field equals attacker

A single unusual field may have an innocent explanation. Correlate it.

Collect everything

More evidence is not automatically better. Collect enough to support or challenge your hypothesis while respecting privacy and scope.

Change the system while investigating

SOC Level 1 analysts should not erase, modify or 'clean up' evidence. Preserve first, then follow the approved escalation/containment process.

Forget the time zone

Record the time zone used in your timeline. Prefer UTC when the assignment does not specify otherwise.

7. Classification language

Use precise language:

Benign: Evidence is consistent with expected behaviour or a normal user mistake.

Suspicious: Evidence contains meaningful anomalies, but available information does not yet prove the controlled incident story.

Confirmed within the lab scenario: Multiple correlated evidence points satisfy the exercise's defined threshold.

Avoid dramatic wording. A professional SOC report explains what happened, what supports the conclusion and what should happen next.

8. Detection improvement thinking

Week 2 also introduces detection engineering at a beginner level. You are not expected to build an advanced production rule. Instead, identify a useful signal that would help analysts notice the same pattern sooner.

A good proposal answers:

- What event fields are required?
- What pattern should trigger attention?
- Over what time window?
- Which normal behaviour might create false positives?
- What severity is appropriate?
- What should the analyst check before escalating?

Example concept: 'Several failed authentications for the same synthetic account followed by a successful authentication within a short lab window.'

The exact thresholds should come from the exercise context rather than being treated as universal production values.

9. Retesting the fixed scenario

A retest is not simply 'the alert disappeared.' Confirm that:

- the intended authentication behaviour still works;
- the previously suspicious/unsafe path no longer behaves the same way;
- useful telemetry still reaches Wazuh;
- the fix did not remove the evidence analysts need;
- your detection proposal still makes sense after the change.

Record the fixed-version evidence separately from the original investigation.

10. Your Week 2 operating sequence

1. Read the Week 2 GitHub Issue.
2. Study this learning pack.
3. Run neolabs login.
4. Run neolabs connect.
5. Confirm Wazuh is receiving your pod telemetry.
6. Investigate only the assigned synthetic accounts/time window.
7. Build the timeline and evidence register.
8. Write the requested report/detection proposal.
9. Retest when the fixed scenario is released.
10. Submit through the central assignment repository Pull Request workflow.

Safety boundary

- Do not attempt to select another pod.

- Do not test real accounts or systems.
- Do not commit Access Codes, session tokens, certificates or private URLs.
- Do not change/delete evidence to make the timeline cleaner.
- If telemetry appears to contain another cohort member's real information, stop and notify a mentor.

Quick knowledge check

- 1 Why is a baseline useful before investigating an unfamiliar login?
- 2 Name four authentication fields that help build a timeline.
- 3 Why is a successful login from an unusual source not automatically proof of compromise?
- 4 What makes evidence reproducible for another analyst?
- 5 What should you verify during a fixed-version retest besides whether an alert disappeared?

Remember

Baseline first. Correlate evidence. State only what the evidence supports.